



Swethank Reddy Akketi

Bachelor of Technology

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SCHOLASTIC ACHIEVEMENTS

- Ranked in the top **2.2%** among 0.18M+ candidates in **JEE Advanced** 2023
- Ranked in the top **0.73%** among 1.1M+ candidates in **JEE Mains** 2023
- Achieved **338th rank** in the **Smart Kid General Knowledge Olympiad** 2016
- Achieved **138th rank** in the **Kwest KMO** 2017
- Attained **55th rank** in the **State-Level English Proficiency Test** by the ELTA 2016-17
- Top **10%** of scorers in the **50th National Mathematics Talent Contest** 2017

KEY PROJECTS

Building Scalable Multi-Agent AI Systems

(Summer 2025)

SOC Project | Web and Coding Club

IIT Bombay

- Engineered **MLP**, **CNN**, **LSTM**, and **Transformer** architectures in **PyTorch** with custom training loops and optimizers, achieving **94.3% accuracy** on MNIST and **89.1%** on IMDB sentiment classification using datasets
- Built **RAG** and **ReAct** agents using **LangGraph**, **LangChain**, and locally hosted LLMs for context-aware reasoning
- Achieved **3.2× inference speed-up** (820ms to 255ms) through **GGUF quantization**, **ONNX export**, and Hugging Face **optimum** optimization for secure, efficient local LLM deployment and low-latency response generation

Gesture Based Text Creation and Recognition

(Summer 2024)

SOC Project | Web and Coding Club

IIT Bombay

- Implemented a **CNN** using **TensorFlow** for handwritten digit recognition on **MNIST**, achieving over **89%** accuracy
- Integrated **MediaPipe** for real-time hand detection and gesture recognition, enabling smooth and easy user interaction
- Designed a real-time gesture-controlled canvas using **OpenCV** for drawing, erasing, resizing, and image manipulation
- Optimized performance and ensured macOS compatibility using **AppleScript** via **osascript**, enhancing interaction

Bulldozer Price Predication using Time series data

(Summer 2024)

Self Project

IIT Bombay

- Developed a **Random Forest Regressor** model to predict bulldozer auction prices using historical **time series** data
- Fine-tuned the model with **hyperparameter tuning** via **RandomizedSearchCV** on a **400K** sample Kaggle dataset
- Implemented thorough data preprocessing and **exploratory data analysis** to enhance prediction accuracy by converting predictor variables to **datetime64** for precise temporal analysis, achieving a validation **R²** score of **0.88**

Regime Detection via Unsupervised Learning from Order Book and Volume Data

(April 2025)

Self Project

IIT Bombay

- Engineered features including **VWAP shifts**, **returns**, and **order book imbalance (OBI)** to capture microstructure signals from high-frequency order book data and extracted features using **correlation analysis** and **heatmaps**
- Applied **K-Means** clustering with **PCA** dimensionality reduction to segment market behavior into **11** distinct regimes
- Achieved **Silhouette Score** of **0.450** and **Davies-Bouldin Index** of **1.227**, demonstrating effective regime separation
- Visualized regime transitions using **Matplotlib** and **Seaborn** to analyze trends, volatility, and behavioral shifts

Real-Time Detection & Recognition with Deep Learning Frameworks

(July 2024)

Learner's Space | Web and Coding Club

IIT Bombay

- Built a real-time **facial recognition system** using **Siamese Neural Networks** to distinguish similar faces
- Applied **YOLOv8** for high-accuracy object detection and classification of vehicles and their components
- Designed and trained deep learning models using **TensorFlow** and **PyTorch**, including **CNNs** for image classification
- Leveraged **OpenCV** for image preprocessing through **filtering**, **transformations**, and **feature extraction**

Conditional Image Synthesis with GANs

(July 2024)

Self Project

IIT Bombay

- Built a **Conditional GAN** in **Keras** to generate images conditioned on class labels, improving specificity and control
- Enhanced the training dataset through **data augmentation** to increase model robustness and generalization
- Conducted **hyperparameter tuning** to refine performance, adjusting learning rates and batch sizes

OTHER PROJECTS

Universal Testing Machine

Guide: Prof. Joseph John | Course Project: Makers Space

(Autumn 2023)

IIT Bombay

- Designed a **Universal Testing Machine** to convert motor rotational motion into linear force for material testing
- Manufactured a **gear box** to increase **torque**, allowing the lead screw to create tension until material failure
- Implemented an **Arduino-based control system** with **sensors** to regulate motor speed for the calculation of force
- Analyzed data to generate **Stress-Strain graphs**, to obtain material properties like **elasticity** and **tensile strength**

IMU Controlled Bot

XLR8 | Electronics and Robotics Club

(September 2023)

IIT Bombay

- Developed an **IMU-driven ESP32 bot** using **ESP-NOW** for real-time wireless control and motion execution
- Processed **gyroscope data** to compute tilt direction and applied **PWM speed control** for smooth motor response
- Integrated **L298N motor driver** with dynamic directional control logic based on **IMU-driven** motion thresholds
- Deployed a **WiFi-based web server** using **ESPAsyncWebServer** for simple and responsive remote command access

TECHNICAL SKILLS

Programming and Web Libraries/Frameworks Software

C++, Python, HTML, CSS, JavaScript, $\text{\LaTeX}$
Node.js, ReactJS, NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, Keras, Django
Arduino, GitHub, AutoCAD, Blender, OpenCV, Adobe Photoshop, Illustrator, Premiere Pro, After Effects

POSITIONS OF RESPONSIBILITY

Creative & Media Convener | Institute Technical Council, IIT Bombay

(April'24-March'25)

Member of a team responsible for publicity of 50+ events to promote Technology culture among 10K+ students

- Coordinated with **20+** Tech-Teams and Tech-Clubs with **60+** members under ITC in aspects of media and publicity
- Led a social media revamp, increasing impressions by **115K+** and significantly boosting outreach by **200%**
- Organising **Tech RnD Expo**, a major event that showcases technical projects of more than **30** technical bodies

College Representative Mentor | Abhyuday, IIT Bombay

(August'24-February'25)

Selected as UG mentor from 66 applicants, working with a group of 13 members to guide students

- Mentored 30+** students from different colleges, guiding them to plan and carry out social initiatives effectively
- Conducted targeted**, sessions to enhance social creativity, by encouraging students in innovative Task/assignments
- Supported** students in **developing** strategies for meaningful social impact , fostering implementation of their jobs

Events & PR Coordinator | Entrepreneurship-Cell, IIT Bombay

(July'24-March'25)

- Ideated the 27th edition of *Eureka!* – Asia's largest Business Model Competition | Prizes worth INR 20M
- Successfully onboarded **10+** Incubators and **3+** Strategic deals to foster growth of **500+** startups in *Eureka!*
- Cemented ties with **Government Bodies, Media Portals, Startup Mentors** increasing *Eureka!*'s reach by **140%**

Creatives Coordinator | Inter IIT Tech Meet 13.0

(September'24-February'25)

Annual technology competition among **19 IITs** involving challenges in coding, robotics, AI, and innovation

- Led media coverage for the **1st** ever offline Pan IIT Tech Board meeting, ensuring extensive engagement
- Enhanced the social media reach by **5x** through strategic content development and targeted audience engagement
- Strengthened event branding by conducting **25+** interviews with participating contingent leaders of IITs

COURSES UNDERTAKEN

Computer Science
Mathematics
Others

Computer Programming and Utilization, Numerical Methods in Civil Engineering
Probability and Statistics, Calculus, Linear Algebra, Differential Equations
Makerspace, Introduction to Entrepreneurship, Economics

EXTRA CURRICULAR ACTIVITIES

Social	Participated in a career counseling campaign, advising over 6,000 students across 60+ schools to enhance their educational and career pathways
Visual Design and Creative Skills	Graphic Design: Proficient in Photoshop and Illustrator for engaging visuals Video Editing: Skilled in Premiere Pro and Final Cut Pro for high-quality edits 3D Modeling and Animation: Proficient in creating detailed 3D assets and animations
Performing Arts	- Performed in Halloween Costume GC representing Hostel-3 as part of a 12-member team - Honored as State Champion in Karate and Kungfu, securing 1st place in the 2013 state-level championship
Miscellaneous	Programming and Web: C++, Python, HTML, CSS, JavaScript - Completed Web Development, Machine Learning & Python for Data Science under Learner's Space IIT Bombay